

Configure Ciphers MACs Kex Algorithms in Nexus Platforms

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Introduction

This document describes the steps to add (or) remove Ciphers, MACs, and Kex Algorithms in Nexus platforms.

Prerequisites

Requirements

Cisco recommends that you understand the basics of Linux and Bash.

Components Used

The information in this document is based on these hardware and software versions:

- Nexus 3000 and 9000 NX-OS 7.0(3)I7(10)
- Nexus 3000 and 9000 NX-OS 9.3(13)
- Nexus 9000 NX-OS 10.2(7)
- Nexus 9000 NX-OS 10.3(5)
- Nexus 7000 NX-OS 8.4(8)
- Nexus 5600 NX-OS 7.3(14)N1(1)

The information in this document was created from the devices in a specific lab environment. All of the devices used in this document started with a cleared (default) configuration. If your network is live, ensure that you understand the potential impact of any command.

Background Information

Some security scans can find weak encryption methods used by Nexus devices. If this happens, changes to the **dcos_sshd_config** file on switches are required to remove insecure algorithms.

Review Available Ciphers, MACs, and Kex Algorithms

To confirm which Ciphers, MACs, and Kex Algorithms a platform uses, check this from an external device. You can use these options:

Option 1. Using CMD Line from PC

Open a CMD line on a PC that can reach the Nexus device and run the command **ssh -vvv**

```
<#root>
```

```
C:\Users\xxxxxx>ssh -vvv <hostname>
```

```
----- snipped -----
```

```
debug2: peer server KEXINIT proposal
```

```
debug2:
```

```
KEX algorithms: diffie-hellman-group1-sha1,diffie-hellman-group14-sha1,diffie-hellman-group-exchange-sha1
```

```
debug2: host key algorithms: ssh-rsa
```

```
debug2: ciphers ctos: aes128-ctr,aes192-ctr,aes256-ctr,aes128-cbc,aes192-cbc,aes256-cbc
```

```
debug2:
```

```
ciphers stoc: aes128-ctr,aes192-ctr,aes256-ctr,aes128-cbc,aes192-cbc,aes256-cbc <--- encryption algo
```

```
debug2: MACs ctos: hmac-sha1
```

```
debug2:
```

MACs stoc: hmac-sha1

<--- mac algorithms

debug2: compression ctos: none,zlib@openssh.com
debug2:

compression stoc: none,zlib@openssh.com <--- compression algorithms

Option 2. Access the "dcos_sshd_config" File by Using Feature Bash-Shell

This applies to:

- N3K running 7. X, 9. X, 10. X
- All N9K codes
- N7K running 8.2 and later

1. Enable the bash-shell feature and in bash mode:

```
switch(config)# feature bash-shell
switch(config)#
switch(config)# run bash
bash-4.3$
```

2. Review the contents from the **dcos_sshd_config** file:

```
bash-4.3$ cat /isan/etc/dcos_sshd_config
```



Note: You can use egrep to look at specific lines: **cat /isan/etc/dcos_sshd_config | grep MAC**

Option 3. Access the "dcos_sshd_config" File by Using a Dplug File

This applies to:

- N3Ks running 6. X that does not have access to the bash-shell
- All N5K and N6K codes
- N7Ks running 6. X and 7. X codes

Steps:

1. Open a TAC case to obtain the dplug file that matches the NXOS version running on the switch.
2. Upload the dplug file to bootflash and create a copy of it.

```
<#root>
```

```
switch# copy bootflash:
```

```
nuova-or-dplug-mzg.7.3.8.N1.1
```

```
bootflash:
```

```
dp
```



Note: A copy ("dp") of the original dplug file is created in bootflash, so, only the copy is removed after the dplug is loaded and the original dplug file remains in bootflash for subsequent runs.

3. Load the copy of the dplug by running the **load** command.

```
<#root>
```

```
n5k-1# load bootflash:dp
Loading plugin version 7.3(8)N1(1)
#####
Warning: debug-plugin is for engineering internal use only!
```

For security reason, plugin image has been deleted.

```
#####
Successfully loaded debug-plugin!!!
Linux(debug)#
Linux(debug)#
```

2. Review the **dcos_sshd_config** file.

```
Linux(debug)# cat /isan/etc/dcos_sshd_config
```

Solution

Step 1. Export the "dcos_sshd_config" File

1. Send a copy of the **dcos_sshd_config** file to bootflash:

```
Linux(debug)# cd /isan/etc/  
Linux(debug)# copy dcos_sshd_config /bootflash/dcos_sshd_config  
Linux(debug)# exit
```

2. Confirm the copy is on bootflash:

```
switch(config)# dir bootflash: | i ssh  
7372 Mar 24 02:24:13 2023 dcos_sshd_config
```

3. Export to a server:

```
switch# copy bootflash: ftp:  
Enter source filename: dcos_sshd_config  
Enter vrf (If no input, current vrf 'default' is considered): management  
Enter hostname for the ftp server: <hostname>  
Enter username: <username>  
Password:  
***** Transfer of file Completed Successfully *****  
Copy complete, now saving to disk (please wait)...  
Copy complete.
```

4. Complete any necessary changes to the file and import back to bootflash.

Step 2. Import the "dcos_sshd_config" File

1. Upload the modified **dcos_sshd_config** file to boot flash.

```
switch# copy ftp: bootflash:  
Enter source filename: dcos_sshd_config_modified.txt  
Enter vrf (If no input, current vrf 'default' is considered): management  
Enter hostname for the ftp server: <hostname>  
Enter username: <username>  
Password:  
***** Transfer of file Completed Successfully *****  
Copy complete, now saving to disk (please wait)...  
Copy complete.  
switch#
```

Step 3. Replace the Original "dcos_sshd_config" File with the Copy

Manual Process (Not Persistent Across Reboots) - All platforms

By replacing the existing **dcos_sshd_config** file under **/isan/etc/** with a modified **dcos_sshd_config** file located in bootflash. This process is not persistent across reboots

1. Upload a modified ssh config file to bootflash:

```
switch# dir bootflash: | i ssh
7372 Mar 24 02:24:13 2023 dcos_sshd_config_modified
```

2. While in bash or Linux (debug)# mode, overwrite the existing **dcos_sshd_config** file with the one in bootflash:

```
bash-4.3$ sudo su
bash-4.3# copy /bootflash/dcos_sshd_config_modified /isan/etc/dcos_sshd_config
```

3. Confirm the changes were successful:

```
bash-4.3$ cat /isan/etc/dcos_sshd_config
```

Automated Process - N7K

By using an EEM script, this is triggered when the log "VDC_MGR-2-VDC_ONLINE" refreshes after a reload. If the EEM is triggered, a py script is run and replaces the existing **dcos_sshd_config** file under **/isan/etc/** with a modified **dcos_sshd_config** file located in bootflash. This only applies to NX-OS versions that support "feature bash-shell".

1. Upload a modified ssh config file to bootflash:

```
<#root>
```

```
switch# dir bootflash: | i ssh
7404 Mar 03 16:10:43 2023
```

```
dcos_sshd_config_modified_7k
```

```
switch#
```

2. Create a py script that applies changes to the **dcos_sshd_config** file. Ensure you save the file with the "py" extension.

```
<#root>

#!/usr/bin/env python
import os
os.system("sudo usermod -s /bin/bash root")
os.system("sudo su -c \"cp

/bootflash/dcos_sshd_config_modified_7

k /isan/etc/dcos_sshd_config\"")
```

3. Upload the Python script to bootflash.

```
<#root>

switch# dir bootflash:///scripts
      175      Mar 03 16:11:01 2023

ssh_workaround_7k.py
```



Note: Python scripts are pretty much the same on all platforms, except for N7K which contains some additional lines to overcome Cisco bug ID [CSCva14865](#).

4. Ensure the **dcos_sshd_config** file name from the script and bootflash (in Step 1.) are the same:

```
<#root>

switch# dir bootflash: | i ssh
      7404      Mar 03 16:10:43 2023

dcos_sshd_config_modified_7k
```

```
switch#
```

```
<#root>

switch# show file bootflash:///

scripts/ssh_workaround_7k.py
```

```
#!/usr/bin/env python
import os
os.system("sudo usermod -s /bin/bash root")
os.system("sudo su -c \"cp /
```

bootflash/dcos_sshd_config_modified_7k

```
/isan/etc/dcos_sshd_config\"")
switch#
```

5. Run the script once, so, **dcos_sshd_config** file is changed.

```
<#root>
```

```
switch#
```

```
source ssh_workaround_7k.py
```

```
switch#
```

6. Configure an EEM script, once the py script is run every time the switch is rebooted, it comes back up.

EEM N7K:

```
<#root>
```

```
event manager applet SSH_workaround
  event syslog pattern "vdc 1 has come online"
  action 1.0 cli command
```

```
"source ssh_workaround_7k.py"
```

```
action 2 syslog priority alerts msg "SSH Workaround implemented"
```



Note: EEM syntax can vary on different NXOS releases (some versions require "action cli" and others "action cli command"), ensure the EEM commands are accurate.

Automated Process - N9K, N3K

1. Upload a modified SSH config file to bootflash.

<#root>

```
switch# dir | i i ssh
7732 Jun 18 16:49:47 2024 dcos_sshd_config
7714 Jun 18 16:54:20 2024
```

dcos_sshd_config_modified

switch#

2. Create a py script that applies changes to the **dcos_sshd_config** file. Ensure you save the file with the "py" extension.

<#root>

```
#!/usr/bin/env python
import os
os.system("sudo su -c \"cp
```

/bootflash/dcos_sshd_config_modified

```
/isan/etc/dcos_sshd_config\"")
```

3. Upload the python script to bootflash.

<#root>

```
switch# dir | i i .py
127 Jun 18 17:21:39 2024
```

ssh_workaround_9k.py

switch#

4. Ensure the **dcos_sshd_config** file name from the script and bootflash (Step 1.) are the same:

<#root>

```
switch# dir | i i ssh
7732 Jun 18 16:49:47 2024 dcos_sshd_config
7714 Jun 18 16:54:20 2024
```

dcos_sshd_config_modified

```
127 Jun 18 17:21:39 2024 ssh_workaround_9k.py
switch#
```

```
<#root>
```

```
switch# sh file bootflash:ssh_workaround_9k.py
#!/usr/bin/env python
import os
os.system("sudo su -c \"cp
```

```
/bootflash/dcos_sshd_config_modified
```

```
 /isan/etc/dcos_sshd_config\"")
switch#
```

5. Run the script once, so, the **dcos_sshd_config** file is changed.

```
<#root>
```

```
switch#
```

```
python bootflash:ssh_workaround_9k.py
```

6. Configure an EEM script, once the py script is ran every time, the switch is rebooted and comes back up.

EEM N9K and N3K:

```
<#root>
```

```
event manager applet SSH_workaround
  event syslog pattern "vdc 1 has come online"
  action 1.0 cli
```

```
python bootflash:ssh_workaround_9k.py
```

```
action 2 syslog priority alerts msg SSH Workaround implemented
```



Note: EEM syntax can vary on different NXOS releases (some versions require "action cli" and others "action cli command"), ensure the EEM commands are accurate.

Automated Process - N5K, N6K

A modified dplug file was created via Cisco bug ID [CSCvr23488](#) to remove these Kex Algorithms:

- diffie-hellman-group-exchange-sha256
- diffie-hellman-group-exchange-sha1
- diffie-hellman-group1-sha1

The dplug files provided via Cisco bug ID [CSCvr23488](#) are not the same as the ones used to access the Linux Shell. Open a TAC case to obtain the modified dplug from Cisco bug ID [CSCvr23488](#).

1. Verify the default **dcos_sshd_config** settings:

```
<#root>
```

```
C:\Users\user>ssh -vvv admin@<hostname>
```

```
---- snipped ----
```

```
debug2: peer server KEXINIT proposal
```

```
debug2:
```

```
KEX algorithms: ecdh-sha2-nistp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffie-hellman-group-exchange-
```

```
<--- kex algorithms
```

```
debug2:
```

```
host key algorithms: ssh-rsa
```

```
debug2: ciphers ctos: aes128-ctr,aes192-ctr,aes256-ctr
```

```
debug2:
```

```
ciphers stoc: aes128-ctr,aes192-ctr,aes256-ctr
```

```
<--- encryption algorithms
```

```
debug2: MACs ctos: hmac-sha1
```

```
debug2:
```

```
MACs stoc: hmac-sha1
```

```
<--- mac algorithms
```

```
debug2: compression ctos: none,zlib@openssh.com
```

```
debug2:
```

```
compression stoc: none,zlib@openssh.com
```

```
<--- compression algorithms
```

2. Create a copy of the modified dplug file.

```
switch# copy bootflash:nuova-or-dplug-mzg.7.3.14.N1.1_CSCvr23488.bin bootflash:dp
```



Note: A copy ("dp") of the original dplug file is created in bootflash, only the copy is removed after the dplug is loaded and the original dplug file remains in bootflash for subsequent runs.

3. Apply the dplug file from Cisco bug ID [CSCvr23488](#) manually:

```
switch# load bootflash:dp2
Loading plugin version 7.3(14)N1(1)
#####
Warning: debug-plugin is for engineering internal use only!
For security reason, plugin image has been deleted.
#####
Successfully loaded debug-plugin!!!
Workaround for CSCvr23488 implemented
switch#
```

4. Verify the new **dcos_sshd_config** settings:

<#root>

```
C:\Users\user>ssh -vvv admin@<hostname>
```

```
---- snipped ----
```

```
debug2: peer server KEXINIT proposal
```

```
debug2:
```

```
KEX algorithms: diffie-hellman-group14-sha1,ecdh-sha2-nistp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521
```

```
debug2: host key algorithms: ssh-rsa
```

```
debug2: ciphers ctos: aes128-ctr,aes192-ctr,aes256-ctr
```

```
debug2:
```

```
ciphers stoc: aes128-ctr,aes192-ctr,aes256-ctr
```

```
debug2: MACs ctos: hmac-sha1
```

```
debug2:
```

```
MACs stoc: hmac-sha1
```

```
debug2: compression ctos: none,zlib@openssh.com
```

```
debug2:
```

```
compression stoc: none,zlib@openssh.com
```

5. Ensure this change is persistent across reboots with an EEM script:

```
event manager applet CSCvr23488_workaround
event syslog pattern "VDC_MGR-2-VDC_ONLINE"
action 1 cli command "copy bootflash:nuova-or-dplug-mzg.7.3.14.N1.1_CSCvr23488.bin bootflash:dp"
action 2 cli command "load bootflash:dp"
action 3 cli command "conf t ; no feature ssh ;feature ssh"
action 4 syslog priority alerts msg "CSCvr23488 Workaround implemented"
```



Note: After the modified dplug is applied, the SSH feature must be reset on this platform. The dplug file must be present in bootflash, and the EEM must be configured with the proper dplug filename. The dplug filename can vary depending on the version of the switch, ensure you modify the script as needed. Action 1 creates a copy of the original dplug file in bootflash to another called "dp", so the original dplug file is not deleted once loaded.

Platform Considerations

N5K/N6K

- MAC (Message Authentication Code) cannot be changed on these platforms by modifying the **dcos_sshd_config** file. The only supported MAC is hmac-sha1.

N7K

- For MACs to be changed, an 8.4 code is required. See Cisco bug ID [CSCwc26065](#) for details.
- "Sudo su" is not available by default on 8.X. Reference Cisco bug ID: [CSCva14865](#). If executed, this error is observed:

<#root>

```
F241.06.24-N7706-1(config)# feature bash-shell
F241.06.24-N7706-1(config)# run bash
bash-4.3$ sudo su
```

Cannot execute /isanboot/bin/nobash: No such file or directory <---

bash-4.3\$

To overcome this, run:

<#root>

```
bash-4.3$
```

```
sudo usermod -s /bin/bash root
```

After you run the prior command, "sudo su" works:

```
bash-4.3$ sudo su
bash-4.3#
```



Note: This change does not survive a reload.

There is a separate **dcos_sshd_config** file for each VDC, if SSH parameters must be modified on a different VDC, ensure you modify the corresponding **dcos_sshd_config** file.

<#root>

```
N7K# run bash
bash-4.3$ cd /isan/etc/
bash-4.3$ ls -la | grep ssh
-rw-rw-r-- 1 root root 7564 Mar 27 13:48
```

dcos_sshd_config

```
<--- VDC 1
-rw-rw-r-- 1 root root 7555 Mar 27 13:48
```

dcos_sshd_config.2

```
<--- VDC 2
-rw-rw-r-- 1 root root 7555 Mar 27 13:48
```

dcos_sshd_config.3

```
<--- VDC 3
```

N9K

- Changes on the **dcos_sshd_config** file is not persistent across reboots on any Nexus platform. If changes must be persistent, an EEM can be used to modify the file every time the switch boots up.
- Enhancements on N9K changes start with 10.4(2) and is also available in 10.5(1). It was not added to newer versions of previous software trains.
- See Cisco bug ID [CSCwd82985](#) for details.

Example CLI from a switch running 10.5(1):

```

switch(config)# ssh ?
cipher-mode      Set Cipher-mode for ssh
ciphers          Ciphers to encrypt the connection <<<<<<<<<
idle-timeout     SSH Client session idle timeout value
kexalgorithms    Key exchange methods that are used to generate per-connection keys <<<<<<<<<
key              Generate SSH Key
keytypes         Public key algorithms that the server can use to authenticate itself to the client
login-attempts   Set maximum login attempts from ssh
login-gracetime  Set login gracetime for ssh connection
macs             Message authentication codes used to detect traffic modification <<<<<<<<<
port             Set port number for ssh
rekey            Renegotiate ssh key

switch(config)# ssh ciphers ?
WORD             Algorithm name to be configured (Max Size 128)
aes256-gcm       <Deprecated> enable aes256-gcm
all              Control known weak SSH algorithms in current version of NX-OS in addition to the b

switch(config)# ssh macs ?
WORD             Algorithm name to be configured (Max Size 128)
all              Control known weak SSH algorithms in current version of NX-OS in addition to the b

switch(config)# ssh kexalgorithms ?
WORD             Algorithm name to be configured (Max Size 128)
all              Control known weak SSH algorithms in current version of NX-OS in addition to the b

```

N7K, N9K (Codes before 10.4 and 10.5), N3K

For platforms running earlier code, you can add Ciphers, MACs, and KexAlgorithms. These versions do not provide the option to select specific algorithms individually; only the 'all' option is supported.

```

N9000(config)# ssh macs ?
all              Enable algorithms supported in current version of SSH

N9000(config)# ssh kexalgorithms ?
all              Enable algorithms supported in current version of SSH
ecdh-sha2-nistp384 Enable ecdh-sha2-nistp384

N9000(config)# ssh ciphers ?
aes256-gcm       Enable aes256-gcm
all              Enable algorithms supported in current version of SSH

```



Note: These commands are available on the Nexus 7000 with releases 8.3(1) and later. For the Nexus 3000/9000 platform, the command becomes available with release 7.0(3)I7(8) and later. (All 9.3(x) releases have this command as well. See [Cisco Nexus 9000 Series NX-OS Security Configuration Guide, Release 9.3\(x\)](#)).
